

Unit 9: Second-Order Linear Equations

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Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

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Unit 9 at a Glance: Core Sub-Topics

A second-order linear equation reads $a(x)y'' + b(x)y' + c(x)y = g(x)$; this unit builds the homogeneous theory and the constant-coefficient solution machine.

- **Existence, uniqueness, superposition** — two side conditions pin a unique solution; linear combinations of solutions are solutions.
- **The Wronskian** — $W(y_1, y_2) = y_1y_2' - y_2y_1'$ tests linear independence and certifies a fundamental set.
- **Abel's identity** — the Wronskian itself solves a first-order ODE, so it is either never zero or identically zero.
- **Reduction of order** — one known solution y_1 manufactures a second, independent y_2 .
- **Constant-coefficient roots** — the characteristic equation $ar^2 + br + c = 0$ splits into real-distinct, repeated, and complex cases.
- **Cauchy–Euler and higher order** — the x^m trick and the degree- n characteristic polynomial.

Part I — Condensed Cheat Sheet

9.1 Introduction to Second-Order Equations

The general linear second-order equation is $a(x)y'' + b(x)y' + c(x)y = g(x)$. It is **homogeneous** when $g(x) = 0$ and **nonhomogeneous** otherwise; its general solution carries *two* arbitrary constants.

Standard and Normalized Forms

$$a(x)y'' + b(x)y' + c(x)y = g(x), \quad y'' + p(x)y' + q(x)y = r(x),$$

with $p = b/a$, $q = c/a$, $r = g/a$ where $a(x) \neq 0$. Linearity means y , y' , y'' each appear to the first power and are never multiplied together.

9.2 Existence, Uniqueness, Superposition, and Independence

On any interval where p , q , and r are continuous, the initial value problem with $y(x_0) = y_0$, $y'(x_0) = y'_0$ has a *unique* solution.

Superposition Principle

If y_1 and y_2 each solve the homogeneous equation $y'' + p y' + q y = 0$, then for *any* constants c_1, c_2 ,

$$y = c_1 y_1 + c_2 y_2$$

is also a solution. The general solution is this two-parameter family precisely when y_1, y_2 are linearly independent.

9.3 Linear Independence and the Wronskian

Two solutions form a **fundamental set** exactly when they are linearly independent, and the Wronskian is the determinant that tests it.

The Wronskian

$$W(y_1, y_2)(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'.$$

If $W(x_0) \neq 0$ at any point, y_1, y_2 are **linearly independent** on the interval and $\{y_1, y_2\}$ is a fundamental set.

9.4 Abel's Identity

Abel's identity says the Wronskian of two solutions of $y'' + p(x)y' + q(x)y = 0$ itself satisfies a first-order ODE, so its sign can never change.

Abel's Formula

For $y'' + p(x)y' + q(x)y = 0$,

$$W(x) = W(x_0) \exp\left(-\int_{x_0}^x p(t) dt\right).$$

Because the exponential is never zero, W is **either identically zero** (dependent solutions) **or never zero** (independent) — there is no middle ground.

9.5 Reduction of Order

Given one nonzero solution y_1 of the homogeneous equation, reduction of order substitutes $y_2 = v(x)y_1$ and solves for v , producing a second independent solution.

Reduction of Order Formula

For $y'' + p(x)y' + q(x)y = 0$ with a known solution y_1 ,

$$y_2 = y_1(x) \int \frac{e^{-\int p(x) dx}}{y_1(x)^2} dx.$$

The substitution $y_2 = vy_1$ collapses the second-order equation to a *first-order* equation in v' , which the integral above solves directly.

9.6 Constant Coefficient Homogeneous Equations

For $ay'' + by' + cy = 0$ with constants a, b, c , try $y = e^{rx}$ to get the **characteristic equation** $ar^2 + br + c = 0$. The discriminant $b^2 - 4ac$ selects one of three cases.

The Three Root Cases

$$ar^2 + br + c = 0, \quad r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- **Real distinct** ($b^2 - 4ac > 0$), roots $r_1 \neq r_2$:

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}.$$

- **Repeated** ($b^2 - 4ac = 0$), one root r :

$$y = (c_1 + c_2 x) e^{rx}.$$

- **Complex conjugate** ($b^2 - 4ac < 0$), $r = \alpha \pm \beta i$:

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x).$$

Common Pitfall: The Repeated-Root Boundary

The repeated case ($b^2 - 4ac = 0$) is the boundary between real and complex roots. Do *not* write $y = (c_1 + c_2)e^{rx}$ — that is a single constant; the second solution carries the factor x , giving $y = (c_1 + c_2x)e^{rx}$. For complex roots, α is the *real* part (in the exponential) and β the *imaginary* part (inside the sine and cosine); swapping them is the classic sign trap.

9.7 The Cauchy–Euler Equation

The Cauchy–Euler (equidimensional) equation $ax^2y'' + bxy' + cy = 0$ has powers of x matching derivative orders. Try $y = x^m$.

Cauchy–Euler Root Cases

Substituting $y = x^m$ gives the indicial equation $am(m - 1) + bm + c = 0$:

- **Distinct** $m_1 \neq m_2$: $y = c_1x^{m_1} + c_2x^{m_2}$.
- **Repeated** m : $y = (c_1 + c_2 \ln x)x^m$.
- **Complex** $m = \alpha \pm \beta i$:

$$y = x^\alpha (c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)).$$

9.8 Higher-Order Linear Equations

An n th-order constant-coefficient equation has a degree- n characteristic polynomial; each real root contributes an exponential, and a root of **multiplicity** k contributes k independent terms.

Repeated Roots at Higher Order

A root r of multiplicity k contributes

$$e^{rx}, xe^{rx}, x^2e^{rx}, \dots, x^{k-1}e^{rx},$$

and a complex pair $\alpha \pm \beta i$ of multiplicity k contributes the analogous $x^j e^{\alpha x} \cos \beta x$ and $x^j e^{\alpha x} \sin \beta x$ for $j = 0, \dots, k - 1$.

Unit 9 Key Takeaway

The general solution of a homogeneous second-order linear equation is $c_1y_1 + c_2y_2$ for any **linearly independent** pair, which the **Wronskian** certifies and **Abel's identity** guards. For constant coefficients the **characteristic roots** dictate the form — **real-distinct**, **repeated** (carry the x), or **complex** ($e^{\alpha x}$ times sines and cosines) — and **reduction of order** recovers a second solution from a first.

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb through the three characteristic-root cases — real distinct, repeated, and complex conjugate — then exercise the Wronskian, Abel's identity, reduction of order, the Cauchy–Euler equation, and a higher-order capstone. Solutions follow in Part III.

1. **(Real distinct roots)** Solve $y'' - 5y' + 6y = 0$. Write the characteristic equation, factor it, and give the general solution.
2. **(Repeated root)** Solve $y'' - 6y' + 9y = 0$. Identify the double root and explain why the factor x appears in the second solution.
3. **(Complex conjugate roots)** Solve $y'' + 4y' + 13y = 0$. Report the complex roots $\alpha \pm \beta i$ and the real general solution.
4. **(Real-root IVP)** Solve the initial value problem $y'' - y' - 2y = 0$, $y(0) = 1$, $y'(0) = 0$.
5. **(Wronskian)** Compute the Wronskian of $y_1 = e^{2x}$ and $y_2 = e^{3x}$, and state what it certifies.
6. **(Abel's identity)** For $y'' + 3y' + 2y = 0$, use Abel's formula to find the Wronskian up to a constant, then verify it directly from the fundamental solutions.
7. **(Reduction of order)** The equation $x^2y'' - 3xy' + 4y = 0$ has the solution $y_1 = x^2$. Use reduction of order to find a second, independent solution y_2 .
8. **(Cauchy–Euler)** Solve $x^2y'' + 2xy' - 6y = 0$ for $x > 0$.
9. **(Repeated-root IVP)** Solve $y'' - 4y' + 4y = 0$, $y(0) = 1$, $y'(0) = 1$.
10. **(Capstone, higher order)** Solve $y''' - 3y'' + 3y' - y = 0$. Identify the multiplicity of the characteristic root and write the general solution.

Part III — Complete Solutions

Solution 1. Try $y = e^{rx}$ to get the characteristic equation, then factor:

$$r^2 - 5r + 6 = (r - 2)(r - 3) = 0 \Rightarrow r_1 = 2, r_2 = 3.$$

The roots are real and distinct, so

$$y = c_1 e^{2x} + c_2 e^{3x}.$$

Solution 2. The characteristic equation is a perfect square:

$$r^2 - 6r + 9 = (r - 3)^2 = 0 \Rightarrow r = 3 \text{ (double)}.$$

A double root supplies only one exponential e^{3x} ; reduction of order furnishes the second solution $x e^{3x}$, so

$$y = (c_1 + c_2 x) e^{3x}.$$

Solution 3. Complete the discriminant:

$$r = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = -2 \pm 3i.$$

With $\alpha = -2$ and $\beta = 3$,

$$y = e^{-2x} (c_1 \cos 3x + c_2 \sin 3x).$$

Solution 4. Characteristic equation $r^2 - r - 2 = (r - 2)(r + 1) = 0$ gives $r = 2, -1$, so $y = c_1 e^{2x} + c_2 e^{-x}$. Apply the initial conditions:

$$y(0) = c_1 + c_2 = 1, \quad y'(0) = 2c_1 - c_2 = 0.$$

Adding the equations, $3c_1 = 1$, so

$$c_1 = \frac{1}{3}, \quad c_2 = \frac{2}{3}, \quad y = \frac{1}{3} e^{2x} + \frac{2}{3} e^{-x}.$$

Solution 5. With $y_1 = e^{2x}$, $y_2 = e^{3x}$:

$$W = y_1 y_2' - y_2 y_1' = e^{2x} \cdot 3e^{3x} - e^{3x} \cdot 2e^{2x} = (3 - 2)e^{5x} = e^{5x}.$$

Since $e^{5x} \neq 0$ for all x , the two solutions are **linearly independent** and $\{e^{2x}, e^{3x}\}$ is a fundamental set.

Solution 6. Here $p(x) = 3$, so Abel's formula gives

$$W(x) = C \exp\left(-\int 3 dx\right) = C e^{-3x}.$$

Check directly: the roots of $r^2 + 3r + 2 = (r + 1)(r + 2) = 0$ are $r = -1, -2$, so $y_1 = e^{-x}$, $y_2 = e^{-2x}$ and

$$W = e^{-x}(-2e^{-2x}) - e^{-2x}(-e^{-x}) = (-2 + 1)e^{-3x} = -e^{-3x},$$

which matches $C e^{-3x}$ with $C = -1$.

Solution 7. Normalize to $y'' - \frac{3}{x}y' + \frac{4}{x^2}y = 0$, so $p(x) = -3/x$ and $e^{-\int p dx} = e^{\int (3/x) dx} = e^{3 \ln x} = x^3$. With $y_1 = x^2$,

$$y_2 = y_1 \int \frac{e^{-\int p dx}}{y_1^2} dx = x^2 \int \frac{x^3}{x^4} dx = x^2 \int \frac{1}{x} dx = x^2 \ln x.$$

The second independent solution is $y_2 = x^2 \ln x$ (the repeated-root Cauchy–Euler form).

Solution 8. Try $y = x^m$ in $x^2 y'' + 2xy' - 6y = 0$. The indicial equation is

$$m(m-1) + 2m - 6 = m^2 + m - 6 = (m+3)(m-2) = 0 \Rightarrow m = 2, -3.$$

The roots are real and distinct, so for $x > 0$

$$y = c_1 x^2 + c_2 x^{-3}.$$

Solution 9. The characteristic equation $r^2 - 4r + 4 = (r-2)^2 = 0$ has the double root $r = 2$, so $y = (c_1 + c_2 x)e^{2x}$. Differentiate: $y' = (c_2 + 2(c_1 + c_2 x))e^{2x}$. Apply the conditions:

$$y(0) = c_1 = 1, \quad y'(0) = c_2 + 2c_1 = 1 \Rightarrow c_2 = -1.$$

Therefore

$$y = (1 - x)e^{2x}.$$

Solution (Capstone). The characteristic equation is a perfect cube:

$$r^3 - 3r^2 + 3r - 1 = (r-1)^3 = 0 \Rightarrow r = 1 \text{ (multiplicity 3)}.$$

A triple root contributes e^x , xe^x , and $x^2 e^x$, so

$$y = (c_1 + c_2 x + c_3 x^2)e^x.$$

Unit 9 Key Takeaway

Every constant-coefficient solution is read straight off the **characteristic roots**: real-distinct roots give $c_1 e^{r_1 x} + c_2 e^{r_2 x}$, a repeated root carries the factor x as $(c_1 + c_2 x)e^{rx}$, and a complex pair gives $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$. The **Wronskian** confirms independence, **Abel's identity** predicts it, and **reduction of order** recovers the missing partner — exactly the $x^2 \ln x$ that the repeated Cauchy–Euler root demands.